

Academic Year 2026-27 (Semester-III)
Data Structures and Algorithms
Question Bank(Units I & II) —

Class: S. Y. B. TECH.

Unit-I: Introduction to Data Structures and Algorithm Analysis

Q. No.	Question Description
1	Define data, information, and knowledge with suitable examples. Also, explain how data structures play a vital role in organizing and processing data efficiently.
2	Apply both pseudo-code and flowchart representations to design an algorithm for finding the largest of three numbers. Compare the two representations.
3	What is an Abstract Data Type (ADT)? Explain the concept with a suitable example.
4	Compare and justify, with real-world examples, the distinction between data structures: 1. linear and non-linear 2. static and dynamic
5	Differentiate between static and dynamic data structures with suitable examples.
6	Define an algorithm. State its essential characteristics and explain with a simple example.
7	What is algorithm complexity analysis? Explain time complexity and space complexity with suitable examples.
8	Explain the Step Count Method of analyzing time complexity with a suitable example.
9	Explain the asymptotic notations used in algorithm analysis: Big-O (O), Theta (Θ), and Omega (Ω) with suitable examples.
10	Explain best case, worst case, and average case time complexity with a suitable example.
11	Explain the concept of sequential organization in data storage with a suitable example.
12	Describe how arrays serve as an Abstract Data Type (ADT) with suitable examples.
13	A 2D array A[4][5] is stored in row-major order with base address 1000 and each element occupying 4 bytes. Calculate the address of element A[2][3]. Also, explain how the address would differ under column-major order.
14	Define an Ordered List. Explain its properties, advantages, and basic operations with examples.
15	Explain how a polynomial can be represented using an array. Illustrate with an example.

Unit-II: Searching and Sorting

Q. No.	Question Description
1	Explain the classification of sorting as internal and external sorting. Also, define the concept of sort stability with examples.
2	Explain the Linear Search algorithm. Write its pseudo-code, analyze its time complexity, and provide a simple example.

Q. No.	Question Description
3	Apply the Sentinel Search technique to search for a given key element in an array of your choice. Write the algorithm and analyze how it reduces the number of comparisons compared to Linear Search.
4	Explain the Binary Search algorithm. With simple example.
5	Explain the Bubble Sort algorithm with a suitable example. Also, state whether it is stable.
6	Explain the adaptive nature of Bubble Sort and discuss its stability. Illustrate with an example how Bubble Sort maintains stability and adapts to already sorted data.
7	Explain the Selection Sort algorithm. Write its pseudo-code or C++ program, and discuss its procedure, time and space complexity, and whether it is stable or not.
8	Write a C++ program for Selection Sort. Comment on the time complexity of Selection Sort.
9	Explain the Insertion Sort algorithm. Write its pseudo-code or C++ program, and discuss its procedure, time and space complexity, and whether it is stable and adaptive.
10	Define a Heap. Explain the Heap data structure, its types, properties, and basic operations with examples.
11	What is a Min Heap and a Max Heap? Sort following elements using Heap Sort in ascending order: 77, 15, 91, 21, 6, 46.
12	What is hashing? Explain the concept and role of hashing in efficient searching with a suitable example.
13	Define hashing. Given a hash table of size 7 and the keys {23, 44, 12, 18, 39, 55}, apply the division-modulo hash function to insert each key. Resolve any collisions using linear probing and show the final hash table.
14	Explain different types of hash functions with suitable examples.
15	What are the different collision resolution techniques in hashing? Explain open hashing or separate chaining with an example.